Where the Rules of Quantum Mechanics Come From

The postulates you were told to accept, read off a single shape

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If you have ever taken a quantum mechanics course, you were handed a list. States live in a Hilbert space. Observables are self-adjoint operators. The state evolves by the Schrödinger equation. The probability of an outcome is the amplitude squared. When you measure, the state collapses. You were told to accept these — not because anyone could tell you *why* they were true, but because, remarkably, they *work*. Every experiment for a hundred years has agreed with them.

A list you must accept and cannot explain is an uncomfortable thing to build a science on. So we asked a smaller, more stubborn question: what if those rules are not rules at all, but features of a shape?

This paper is our answer. On a single geometric object — a five-dimensional curved space called D_{IV}^5 — the list stops being postulates and becomes things the shape *does*. We are going to walk you through it: what we claim, how it works, and — the part we care about most — exactly how you can check us. We will tell you the tier of every claim as we go: derived when we have proved the mechanism, identified when the match is strong but the mechanism is not yet nailed, and so on. When we do not yet know something, we will say so plainly. That honesty is not a disclaimer bolted on at the end; it is the whole point.

What we claim

Quantum mechanics has, if you count carefully, about ten load-bearing ideas — the five postulates above, plus the structural companions every course teaches alongside them: the uncertainty principle, the arrow of time, how you combine two systems, the spacetime facts (Lorentz symmetry, CPT), and the rule that matter is made of fermions.

The claim is this: all ten come off the one shape, with nothing added by hand beyond the shape itself — and we name the shape's own inputs plainly in a moment, so "nothing added by hand" means *no free knobs*, not *no assumptions*. Not "we can fit them" — *come off*. The space has its own natural space of functions to live in, its own natural clock to evolve by, its own natural notion of size to measure with. When you write those down, you are writing down quantum mechanics.

Ten of ten, at a stated tier each, with two honest caveats we will name. That is a stronger and more checkable claim than "quantum mechanics is derived" — and, we will argue, it is the true one.

How it works

Here is the shape, in one breath. D_{IV}^5 is a *bounded symmetric domain* — think of it as a particular curved, five-complex-dimensional generalization of the interior of a disk, with a great deal of symmetry. It is not a shape we chose to make quantum mechanics appear; it is forced by a short list of demands laid out elsewhere in this program, and five whole numbers fall out of it (rank 2, and $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$). Everything below is that one object, examined from different sides.

States. A curved space like this has a canonical space of "nice" functions living on it — its Bergman/Hardy space. That space is a complex Hilbert space, automatically. You do not posit the Hilbert space of quantum mechanics; it is the shape's own function space. *(Derived.)*

Observables and evolution. The shape has a most-symmetric motion — a preferred way it can be gently rotated — generated by one operator. The unitary face of that one generator *is* Schrödinger evolution, and the physical observables are functions of it. Time is not smuggled in; it is the shape's own clock, the parameter along that motion. *(Derived.)*

The Born rule. To do probability you need a measure — a notion of "how much" of the space sits where. A symmetric space has exactly one measure that respects all its symmetries. Demand that the probabilities not depend on how you happen to draw your coordinates, and that unique measure is forced; its density is $|\text{amplitude}|^2$. The famous square is the shape insisting on its own symmetry. *(Derived.)*

Uncertainty. Why can you not know position and momentum together? Because the shape is *curved*, and you are trying to image a discrete interior onto a smooth continuum. The unavoidable blur of that imaging is the curvature of the space — a quantity a geometer would call the holomorphic sectional curvature — and it sets the $\hbar/2$ floor. This one we have *proved* as a theorem; the one honest caveat is that the exact number out front depends on a curvature convention we pin to the standard references rather than assert from memory. *(Proved, with a named normalization caveat.)*

The arrow of time. Evolution has a *forward* face too — a one-way contraction, a heat flow — and its positivity is what makes time irreversible. The arrow is not an extra law; it is the sign of one thing. *(Derived.)*

Combining systems, and measurement. Two particles live on the shape-times-the-shape, and the natural object there gives the tensor product — cleanly, for *distinguishable* subsystems (that scope is our second honest caveat). Measurement is the shape committing: a forced, one-way step that makes an outcome definite, with the odds coming out Born. We derive the *process* and the *odds*. What we do not derive — and want to be loud about — is *which* single outcome you get on a given run. That specific result is a boundary datum, given, exactly as every physical theory takes its initial data as given. We have not "solved measurement," and we will not pretend to. *(Process and odds derived; the single outcome is physical and irreducible.)*

Spacetime, and CPT. The descent from the shape's symmetry group down to ordinary $3 + 1$ spacetime forces the signature — three space, one time — rather than assuming it. CPT holds too, but here we are careful: CPT is *universal*, true of essentially any sensible relativistic theory, so its holding is inherited, not a distinctive triumph of ours. *(Signature structure-derived; CPT universal, not distinctive.)*

Why matter is fermionic. This is the axiom we are proudest of, and the reason is not that it was easy — it is that it was *hard*, and we refused to fake it. For three separate rounds we held this one as an honest posit, because the pretty story we had (matter as a kind of superpartner of the geometry) could not be *forced* — it was interpretation, and we would not bank interpretation. What finally closed it was cleaner and did not need the pretty story at all: a committed record is an idempotent — do it twice, get the same thing — and an idempotent count can only be 0 or 1. That "0 or 1" is exactly Pauli's exclusion; it is a *bit*. So a lasting record is fermionic, while the bosonic things $(0, 1, 2, \ldots)$ can only be mediators, never records. In three-plus-one dimensions there is no third option. A record is a bit, and a bit is a fermion. *(Derived — and stronger for having been refused three times before it earned it.)*

What we assume — stated plainly

BST is not a theory from nothing, and we will not pretend it is. Here is the whole floor, benchmarked honestly:

- One geometric premise: spacetime is the shape's conformal descent. And note — the $3 + 1$ signature is *structure-derived* from this, not assumed. That is a genuine shortening: general relativity *takes* the signature; we force its structure.
- One dimensionful scale: the electron mass, measured to three parts in ten billion, non-gravitationally. It plays the role Newton's constant plays in general relativity — one anchor that sets the scale of everything. (From it, as it happens, the geometry predicts Newton's constant itself to about 0.07% — a genuine prediction, with both the fine-structure constant and the electron mass taken as measured.)

- Boundary data: the initial conditions and the single measurement outcomes — the things every theory takes as given.

Against that floor, the *entire dimensionless structure of quantum mechanics is forced with zero free dimensionless parameters.* The number that makes that a win is the Standard Model's roughly nineteen dimensionless inputs, which we replace with none — not general relativity, which carries no dimensionless coupling either and so ties us at zero. So the honest comparison is: a floor no longer than general relativity's — one scale, and a signature we derive rather than assume — and zero dimensionless freedom where the Standard Model spends nineteen. The floor is a strength, not an apology.

How to confirm it — please do

We mean this literally. The entire program is public. Every claim above carries a tier, and every number has a short program — we call them *toys* — that checks it, sitting in the repository next to the text. You do not have to trust the list as a whole. Pick the row you are most skeptical of — the Born rule, say, or the fermion argument — and check that one. Run the toy. Read the theorem it cites. If a single row fails, you have learned something real, and so have we. A claim you can check one row at a time, without swallowing the rest, is a different and more honest thing than a claim you must take whole.

The math is on GitHub. Fork it. Break it if you can.

What this is not

- It is not "measurement is solved." The process and the odds are derived; the single outcome you actually get is physical, given, irreducible. We refuse the over-claim that quantum randomness has been explained away.
- It is not "CPT is our discovery." CPT is universal; among these axioms, the genuinely *distinctive* one is uncertainty — the curvature.
- It is not "no caveats." Two travel with the scorecard: the exact curvature normalization for uncertainty, and the distinguishable-particle scope of the tensor product. We mark them rather than paper over them.

The claim, exactly

The axioms of quantum mechanics are features of one shape — ten of ten accounted for (derived, proved, or structure-derived, each at its stated tier, with no posit left standing), on a floor no longer than general relativity's, with two support-axiom caveats named, and the hardest axiom, *why matter is fermionic*, closed by a forced computation only after we refused it three times. That is a stronger and more checkable statement than "quantum mechanics is derived," and — as far as we can tell, and we have tried hard to break it — it is the true one.

Notes for the team (not for the reader): NARRATIVE pilot, drafted by Keeper in the Guide voice, holding Cal's §456 tier-inflation checklist — CPT kept universal-not-distinctive; uncertainty Proved with the curvature caveat; measurement-not-solved explicit; the refused-three-times history kept; "no longer than GR / one scale" not "fewer than any theory"; "zero dimensionless free parameters" scoped to the SM contrast (GR ties); G stated as a 0.07% prediction with both inputs measured. Owes: Cal tier-inflation re-read of THIS voice (the warm register is exactly where a tier quietly inflates); the separate standard-academic version; Casey GO. The accurate technical core is `BST_paper_Axioms_of_QM_from_D_IV5_DRAFT`*. Nothing external without both voices + Cal-vet + Casey GO.*